



IT SIMPLERA INSTITUTE INTERNSHIP PROGRAM

WEEK 02 REPORT

pfSense Firewall Deployment and Configuration

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1. Introduction

A firewall is one of the most important components of network security, acting as the first line of defense against unauthorized access and malicious traffic. pfSense is a powerful open-source firewall and router platform that provides advanced security features such as packet filtering, VPN support, intrusion prevention, and traffic monitoring.

In this project, pfSense Firewall was deployed in a virtual environment. The firewall was configured with pfBlockerNG, GeoIP-based country blocking, DNSBL website filtering, administrator access control, and firewall log monitoring to create a secure network environment.

2. Objective

The objectives of this project were:

- Deploy the pfSense Firewall virtual appliance.
 - Configure basic firewall settings.
 - Install and configure the pfBlockerNG package.
 - Implement GeoIP-based country blocking.
 - Configure DNSBL to block specified websites.
 - Secure administrator access to the pfSense WebGUI.
 - Monitor firewall logs and reports.
 - Validate security policies through practical testing.
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3. Software Requirements

- VMware Workstation
 - pfSense Firewall
 - Windows 10 Virtual Machine
 - Internet Connection
 - Web Browser
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4. pfSense Firewall Deployment

The pfSense virtual appliance was successfully installed and configured inside VMware Workstation. The virtual machine was assigned WAN and LAN interfaces, and the initial installation wizard was completed successfully.

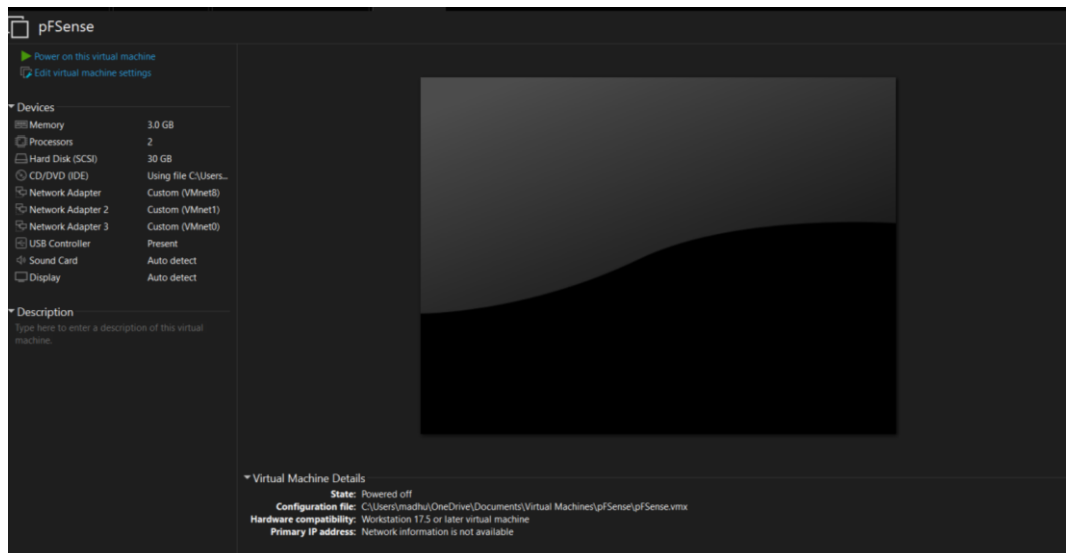


Figure 1 pfSense Firewall Installation Completed

5. Initial Firewall Configuration

After installation, the pfSense WebGUI was accessed using the LAN IP address. The initial setup wizard was completed by configuring the hostname, DNS settings, timezone, administrator password, and network interfaces.

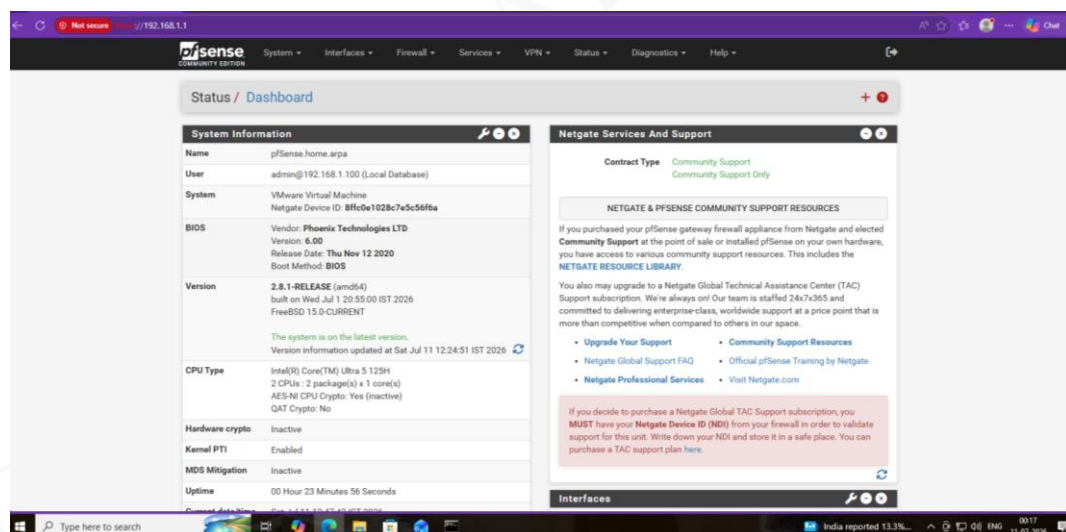


Figure 2 pfSense Dashboard

6. Installing pfBlockerNG

The pfBlockerNG package was installed using the pfSense Package Manager. After installation, the package was enabled to provide IP filtering, DNS-based blocking, and GeoIP filtering capabilities.

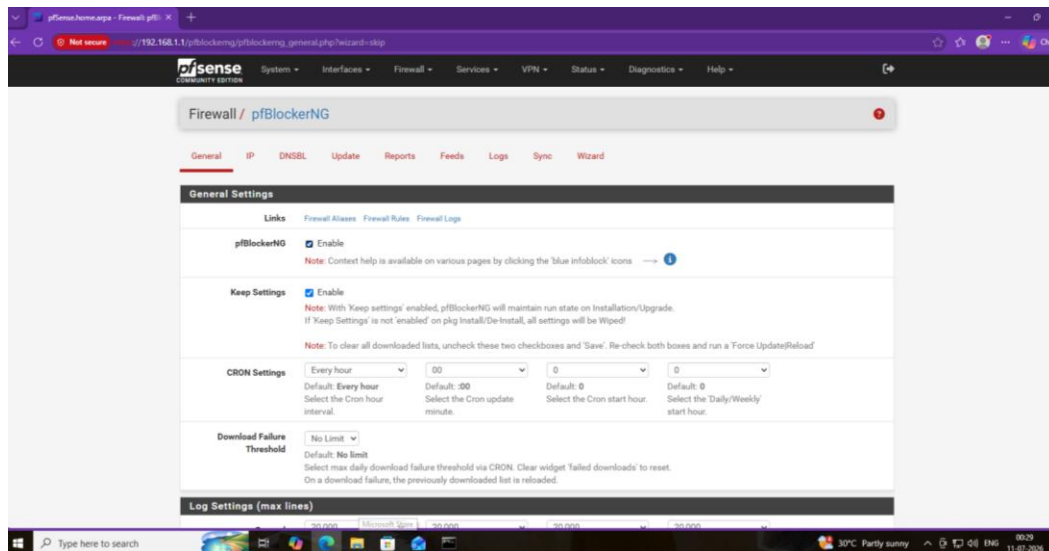


Figure 3 pfBlockerNG Installation

7. GeoIP Country Blocking

GeoIP filtering was configured to block traffic originating from selected countries. Country block lists were created and applied using pfBlockerNG IP rules to enhance network security.

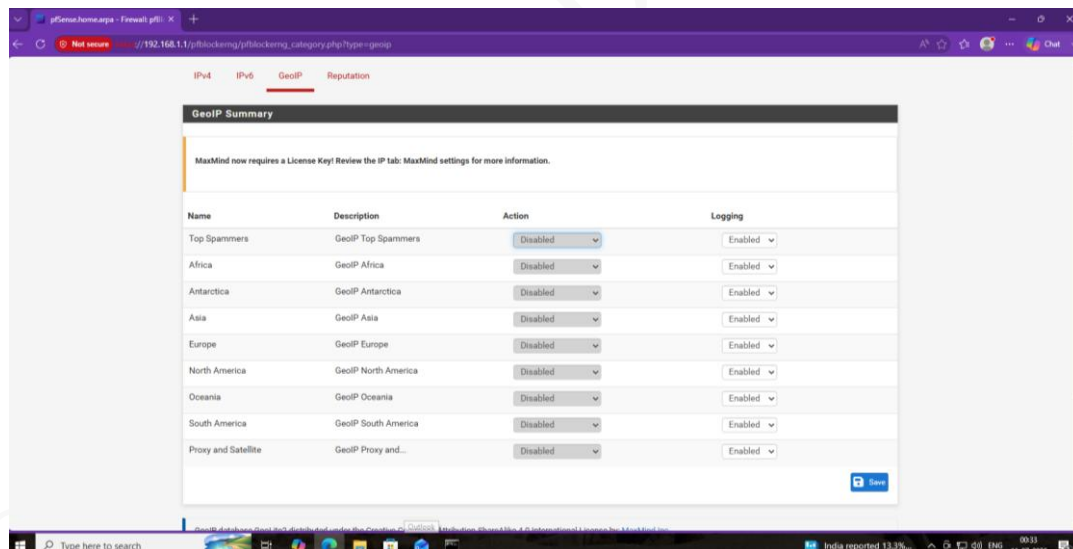


Figure 4 GeoIP Configuration

8. DNSBL Website Filtering

DNSBL (DNS Blacklist) was configured to block access to specified websites such as Facebook and Chat.com. The custom blacklist was applied successfully through pfBlockerNG.

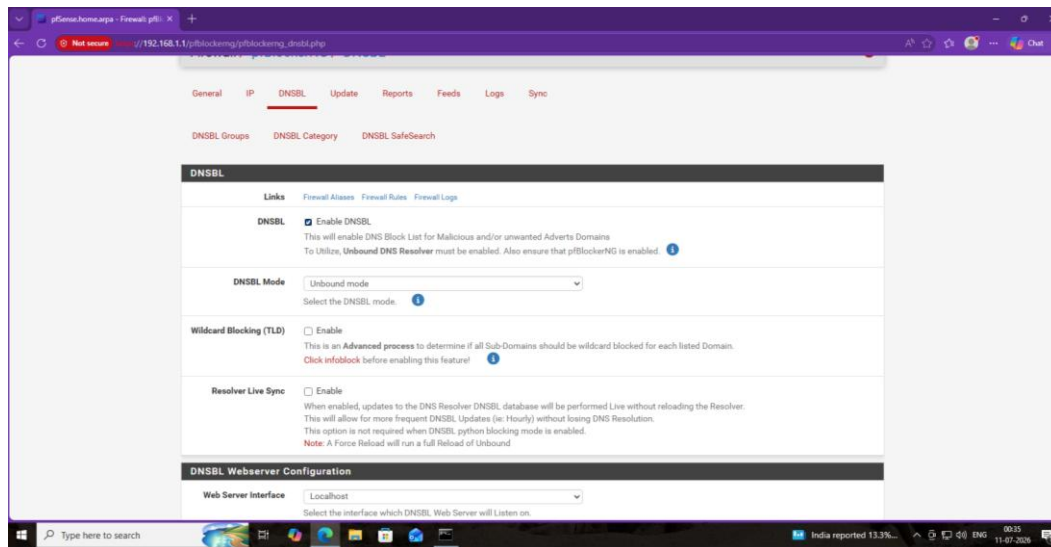


Figure 5 DNSBL Configuration

9. Website Blocking Validation

Website filtering was validated by attempting to access blocked websites from a client machine. The firewall successfully prevented access, confirming that DNSBL was functioning correctly.

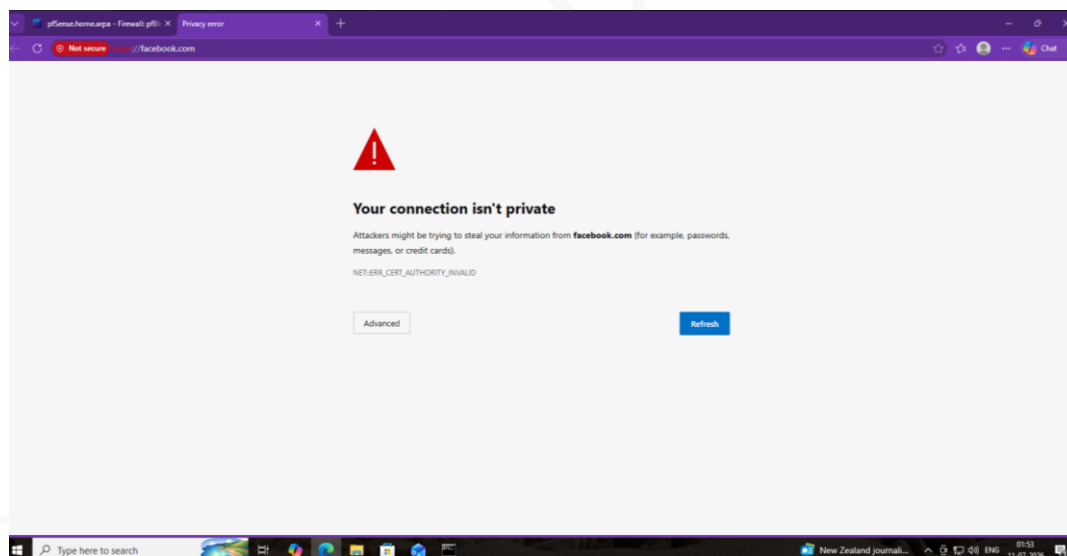


Figure 6 Website Blocking Test

10. Administrator Access Control

Firewall rules were created to restrict administrative access to the pfSense WebGUI. Only authorized IP addresses were permitted to access the management interface while unauthorized connections were blocked.

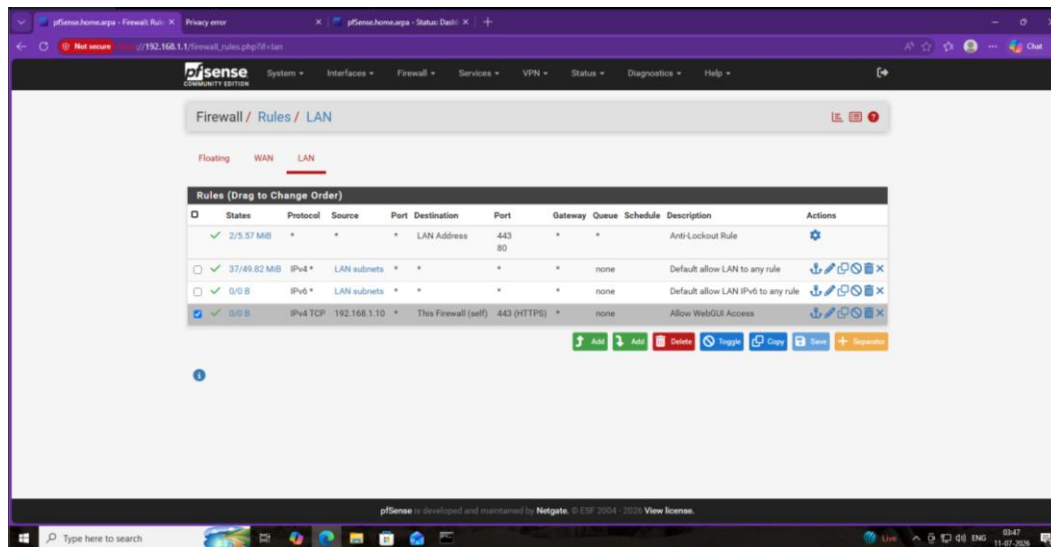


Figure 7 Firewall Rules

11. Firewall Log Monitoring

Firewall activity was monitored using the pfSense System Logs. Both allowed and blocked traffic entries were reviewed to verify that firewall policies were functioning correctly.

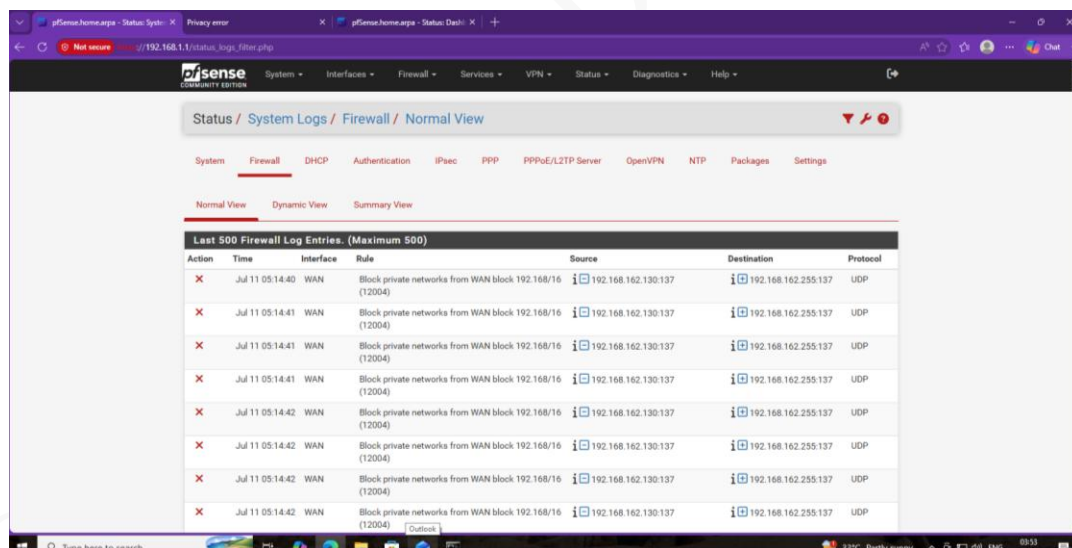


Figure 8 Firewall Logs

12. pfBlockerNG Reports

The pfBlockerNG reporting dashboard was used to analyze blocked traffic generated by GeoIP filtering and DNSBL. The reports confirmed that the configured security policies were actively filtering network traffic.

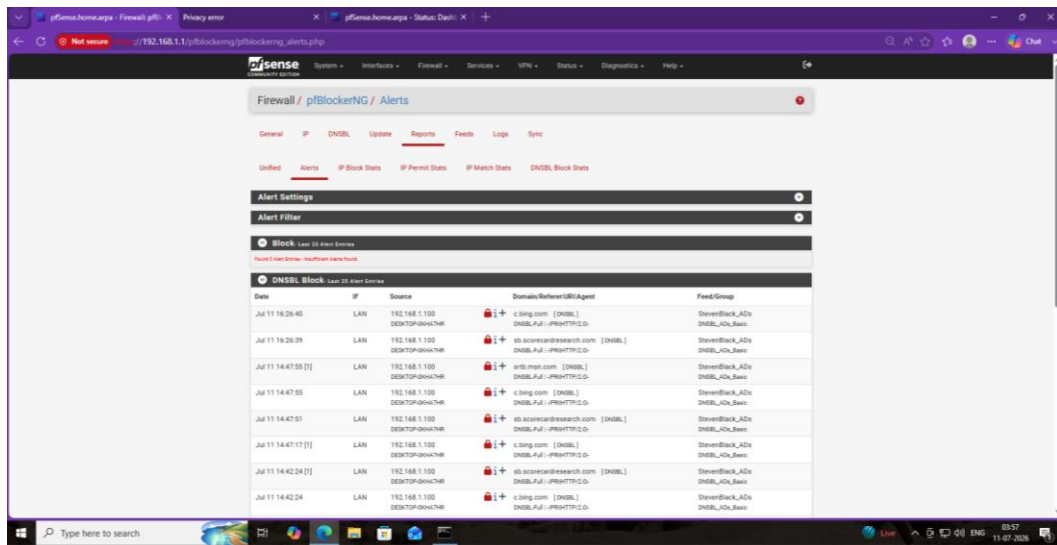


Figure 9 pfBlockerNG Reports

13. Results

The pfSense Firewall was successfully deployed and configured to provide centralized network security. pfBlockerNG was installed, GeoIP filtering and DNSBL website blocking were implemented, administrator access was secured, and firewall logs confirmed the successful enforcement of all configured security policies.

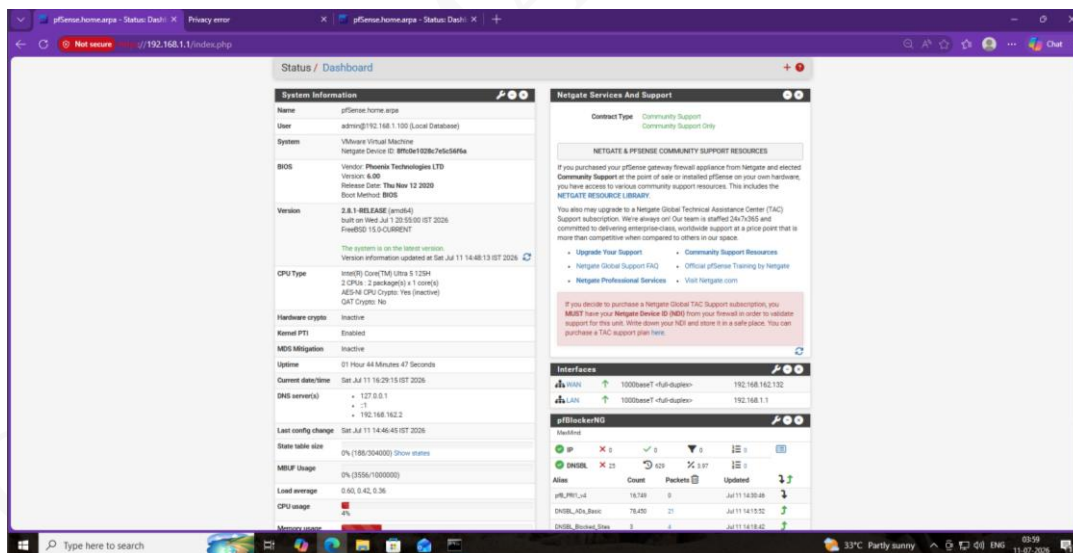


Figure 10 Final pfSense Dashboard

14. Learning Outcomes

Through this project, I learned:

- pfSense Firewall deployment and configuration

- Network firewall administration
 - Installation and configuration of pfBlockerNG
 - GeoIP-based country blocking
 - DNSBL website filtering
 - Firewall rule creation
 - Administrator access control
 - Firewall log analysis
 - Network traffic monitoring
 - Practical Blue Team security operations
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15. Conclusion

This project provided practical experience in deploying and managing a pfSense Firewall within a virtual environment. By implementing GeoIP filtering, DNSBL website blocking, administrator access control, and firewall monitoring, I gained hands-on knowledge of network security and firewall administration. The project strengthened my understanding of traffic filtering, security policy enforcement, and monitoring techniques commonly used in enterprise Security Operations Centers (SOCs).
